

Submitted By:

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Course:

Diploma in Artificial Intelligence and Machine Learning

Report Name:

Diabetes Prediction Using Machine Learning.

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Abstract

Diabetes is one of the most common chronic diseases affecting millions of people worldwide. Early diagnosis can reduce health complications and improve treatment outcomes. This project develops a machine learning model that predicts whether a patient is diabetic based on medical attributes such as glucose level, blood pressure, insulin, BMI, age, and other health indicators. Logistic Regression is used as the prediction algorithm after preprocessing the dataset. The model provides satisfactory prediction accuracy and demonstrates how machine learning can support healthcare professionals in early diagnosis.

Introduction

Machine Learning is transforming the healthcare industry by enabling intelligent disease prediction systems. Diabetes is a chronic disease that requires timely diagnosis to avoid severe complications. Predicting diabetes using patient health records helps doctors make faster and more accurate decisions.

This project uses supervised machine learning techniques to classify patients as diabetic or non-diabetic based on medical information collected from the dataset.

Problem Statement

Many diabetes cases remain undiagnosed until serious complications occur. Manual diagnosis can be time-consuming and may depend on multiple clinical factors. Therefore, there is a need for a prediction system that assists healthcare professionals by providing quick and reliable predictions using patient medical data.

Dataset Description

The dataset contains medical information collected from patients.

Input Features:

1. Pregnancies
2. Glucose
3. Blood Pressure
4. Skin Thickness
5. Insulin
6. BMI
7. Diabetes Pedigree Function
8. Age

Output Feature:

□ Outcome (0 = Non-Diabetic, 1 = Diabetic)

Technologies Used

Programming Language:

- Python

Libraries:

1. NumPy
2. Pandas
3. Matplotlib
4. Seaborn
5. Scikit-learn
6. Pickle

Development Environment:

- Google Colab
- Jupyter Notebook

Methodology

The project follows these steps:

1. Import required libraries.
2. Load the dataset.
3. Explore the dataset.
4. Perform preprocessing.
5. Separate input and output variables.
6. Split the data into training and testing sets.
7. Apply feature scaling.
8. Train the Logistic Regression model.
9. Evaluate the model.
10. Predict diabetes for new patient data.

Data Preprocessing

The dataset was examined for missing values and statistical information. Features were separated from the target variable. The data was divided into training and testing datasets. StandardScaler was used to normalize the input features before training the model.

Model Building

Logistic Regression was selected because it performs well for binary classification problems.

Training Process:

- Training Dataset: 80%
- Testing Dataset: 20%
- Random State: 42

The model learned the relationship between medical parameters and diabetes status during training.

Results

The model successfully predicted diabetes using patient medical records.

Evaluation metrics:

- Training Accuracy
- Testing Accuracy
- Confusion Matrix
- Classification Report

The confusion matrix showed correct classification of most patients, while the classification report provided precision, recall, and F1-score values.

Advantages

- Fast prediction
- Easy to use
- Supports early diagnosis
- Low computational cost
- Suitable for healthcare applications

Limitations

- Accuracy depends on data quality.
- Only one machine learning algorithm is used.
- Cannot replace professional medical diagnosis.
- Performance may vary with different datasets.

Future Scope

- Deploy as a web application.
- Develop a mobile application.
- Compare multiple machine learning algorithms.
- Increase dataset size for better accuracy.
- Integrate with hospital management systems.

Conclusion

The Diabetes Prediction System was successfully developed using Logistic Regression. Data preprocessing, feature scaling, and model training resulted in reliable prediction performance. The project demonstrates the practical use of machine learning in healthcare and highlights its potential to assist doctors in early diagnosis and decision-making.

References

1. Scikit-learn Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. UCI Machine Learning Repository
7. Kaggle Diabetes Dataset